



Topics to cover

1. Response model
2. Status code
3. Clean API contracts

① Why Response Models?

```
return {  
    "message": "User created",  
    "user": user  
}
```

Problem:

- ⊙ Response shape is not strictly defined
- ⊙ Docs are less clear
- ⊙ Frontend doesn't know exact response format guarantee

With the help of Response models:

- ⊙ You define contract
- ⊙ API becomes Predictable
- ⊙ Docs become clear
- ⊙ Safer for Frontend & other services

② Create a Response Model

```
from pydantic import BaseModel  
class UserCreate(BaseModel):
```

```
    name: str
```

```
    age: int
```

```
    email: str
```

```
class UserResponse(BaseModel):
```

```
    id: int
```

```
    name: str
```

```
    age: int
```

```
    email: str
```




③ Use Response Model in Route

```
from fastapi import APIRouter, status
from app.model.user import UserCreate, UserResponse
```

```
router = APIRouter(prefix="/users",
                    tags=["users"])
```

```
@router.post("/", response_model=
    UserResponse, status_code=status.HTTP_
    201_CREATED)
```

```
def create_user(user: UserCreate):
```

```
    save_user = {
        "id": 1,
        "name": user.name,
        "age": user.age,
        "email": user.email
    }
```

```
    return save_user
```

④ What FastAPI Does for you

① validate request using UserCreate

② validate response using UserResponse

③ Filter out extra field if any

④ shows clean schema in /docs

⑤ sends 201 created status code



⑤ Status code

Instead of always returning 200:

- ① 201 Created → When resource is created
- ② 200 OK → Normal Success
- ③ 400 Bad Request → Client error
- ④ 404 NOT Found → Resource not Found
- ⑤ 401 Unauthorized → Not Logged in

In FastAPI:

```
from fastapi import status
```

```
status_code = status.HTTP_201_CREATED
```

⑥ Real - World Analogy

Think of API like a contract:

- ① Request Model = This is form client must fill
- ② Response Model = This is exactly what server promises to return.
- ③ status code = Approved, Rejected, created, etc.